

Piecewise Self-Similar Radiation-Hydrodynamics Analytic Profiles

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1 Wave Front and Mass Coordinates

All physical quantities are scaled using nanoseconds (ns) for time, g/cm^2 for mass coordinates, g/cm^3 for density, MBar for pressure, km/s for velocity, and HeV for temperature.

Exact physical constants (according to SI):

Quantity	value	SI units
c	299792458	m / s
k_B	1.380649×10^{-23}	J / K
q	$1.602176634 \times 10^{-19}$	J
h	$6.62607015 \times 10^{-34}$	J s

Other fundamental quantities derived are:

Quantity	Theoretical Expression (SI)	CGS Units
HeV	$\frac{q}{k_B} \cdot 100$	K
σ_{SB}	$\frac{2\pi^5 k_B^4}{15c^2 h^3}$	$\text{erg}/(\text{cm}^2 \cdot \text{s} \cdot \text{K}^4)$
a	$\frac{4\sigma_{SB}}{c}$	$\text{erg}/(\text{cm}^3 \cdot \text{K}^4)$

1.1 Case 1: $\tau = 0$

Material model.

Quantity	value	CGS units
f	3.4×10^{13}	erg / g
f_{Kelvin}	$\frac{f}{\text{HeV}^\beta}$	erg / g
β	1.6	-
μ	0.14	
g	7200	cm^2 / g
g_{Kelvin}	$\frac{g}{\text{HeV}^\alpha}$	erg / g
α	1.5	-
λ	0.2	-
γ	1.25	-
r	0.25	-
ρ_0	19.32	g/cm^3

1.2 Self Similar Constants

$$A = T_0(rf)^{\frac{1}{\beta}} = 120472196.47695 \quad (1)$$

$$B = \frac{16\sigma g}{3\beta(rf)^{\frac{4+\alpha}{\beta}}} = 1.70920e - 28 \quad (2)$$

$$n = \frac{4 + \alpha}{\beta} = \frac{55}{16} \quad (3)$$

$$k = 1 - \mu = 1 - 0.14 = 0.86 \quad (4)$$

$$a_1 = \frac{\mu}{\beta} = \frac{7}{80} \quad (5)$$

$$a_2 = \frac{2 - \mu}{\beta} = \frac{93}{80} \quad (6)$$

$$a_3 = -\frac{2}{\beta} - \tau = -\frac{5}{4} \quad (7)$$

$$b_1 = \frac{\beta(2 + \lambda) - \mu(4 + \alpha)}{\beta} = \frac{55}{32} \quad (8)$$

$$b_2 = \frac{2\beta - \beta\lambda - 8 - 2\alpha + \mu(4 + \alpha)}{\beta} = -\frac{723}{160} \quad (9)$$

$$b_3 = \frac{8 - 3\beta + 2\alpha}{\beta} = \frac{31}{8} \quad (10)$$

$$a = \frac{2\beta - \beta\lambda - 8 - 2\alpha + \mu(4 + \alpha)}{4 + 2\lambda - 4\mu} = -\frac{245}{128} \quad (11)$$

$$b = \frac{\mu - 2}{4 + 2\lambda - 4\mu} = -\frac{31}{64} \quad (12)$$

$$c = \frac{3\mu - 2\lambda - 2 + \tau(2(\beta - \alpha - 4) - \beta\lambda + \mu(4 + \alpha))}{4 + 2\lambda - 4\mu} = -\frac{33}{64} \quad (13)$$

$$a_v = \frac{2\beta - 4 - \alpha}{2 + \lambda - 2\mu} = -\frac{115}{96} \quad (14)$$

$$b_v = \frac{-1}{2 + \lambda - 2\mu} = -\frac{25}{48} \quad (15)$$

$$c_v = \frac{1 - \tau(4 + \alpha - 2\beta)}{2 + \lambda - 2\mu} = \frac{25}{48} \quad (16)$$

$$a_u = \frac{\beta(2 + \lambda) - \mu(4 + \alpha)}{4 + 2\lambda - 4\mu} = \frac{275}{384} \quad (17)$$

$$b_u = \frac{-mu}{4 + 2\lambda - 4\mu} = -\frac{7}{192} \quad (18)$$

$$c_u = \frac{\tau(\beta(2+\lambda) - \mu(4+\alpha)) + \mu}{4+2\lambda-4\mu} = \frac{7}{192} \quad (19)$$

$$a_p = \frac{\beta\lambda + 4 + \alpha - \mu(4+\alpha)}{2+\lambda-2\mu} = \frac{505}{192} \quad (20)$$

$$b_p = \frac{1-\mu}{2+\lambda-2\mu} = \frac{43}{96} \quad (21)$$

$$c_p = \frac{\tau(\beta\lambda + 4 + \alpha - \mu(4+\alpha)) + \mu - 1}{2+\lambda-2\mu} = -\frac{43}{96} \quad (22)$$

1.3 Wave Coordinate Normalization Variables

The ablated mass coordinate $m_f(t)$ and shocked mass coordinate $m_s(t)$ grow as exact power laws in time:

$$m_f(t) = \xi_f A^{-a} B^{-b} t^{-c} \approx 0.00102 \left(\frac{t}{\text{ns}} \right)^{\frac{33}{64}} \text{ g/cm}^2 \quad (23)$$

$$m_s(t) = m_{s0} \left(\frac{t}{\text{ns}} \right)^{\frac{149}{192}} \approx 0.01209 \left(\frac{t}{\text{ns}} \right)^{\frac{149}{192}} \text{ g/cm}^2 \quad (24)$$

where ξ_s is the shock similarity front coordinate:

$$\xi_s = 1.67133 \quad (25)$$

So we can define the normalized mass coordinates $y = m/m_f(t)$ and $y_s = m/m_s(t)$ directly as:

$$y = \frac{m}{m_f(t)} \approx 981.45247 \left(\frac{m}{\text{g/cm}^2} \right) \left(\frac{t}{\text{ns}} \right)^{-\frac{33}{64}} \quad (26)$$

$$y_s = \frac{m}{m_s(t)} \approx 82.68924 \left(\frac{m}{\text{g/cm}^2} \right) \left(\frac{t}{\text{ns}} \right)^{-\frac{149}{192}} \quad (27)$$

2 Explicit Subsonic Ablation Profiles ($m < m_f(t)$)

2.1 Fluid Velocity $u(m, t)$

1. **Dimensionless Fit Function:** Using the optimal Candidate 12 (Asymptotic Blend 10-Parameter 4-Power model):

$$U(y) \approx W(y)U_{\text{left}}(y) + (1 - W(y))U_{\text{right}}(y)$$

where:

$$U_{\text{left}}(y) = -8.36010 + 33.08430y^{0.69118} - 50.00000y^{20.69118} - 50.00000y^{30.69118} - 50.00000y^{40.69118}$$

$$U_{\text{right}}(y) = -0.00002 - 3.51428(1-y) + 10.16871(1-y)^2 - 20.00000(1-y)^3 + 14.76371(1-y)^4$$

$$W(y) = \frac{1-y}{1+(y/0.20621)^4}$$

2. **Theoretical Scaling:**

$$u(m, t) = U(y) A^{a_u} B^{b_u} t^{c_u} = U(y) A^{\frac{275}{384}} B^{-\frac{7}{192}} t^{\frac{7}{192}}$$

3. **Parametrized Shape:**

$$u(m, t) \approx U(y) A^{\frac{275}{384}} B^{-\frac{7}{192}} t^{\frac{7}{192}}$$

4. **Substitution of scaling constants and units:** Using physical conversions ($u_{\text{scale}} \approx 29.60156 \text{ km/s}$):

$$u(m, t) \approx U(y) \cdot 29.60156 \left(\frac{t}{\text{ns}} \right)^{\frac{7}{192}} \text{ km/s}$$

5. **Fully Substituted CGS function:** Substituting $y \approx 981.45247 \left(\frac{m}{\text{g/cm}^2} \right) \left(\frac{t}{\text{ns}} \right)^{-\frac{33}{64}}$:

$$u(m, t) \approx [W(y)u_{\text{left}}(y) + (1 - W(y))u_{\text{right}}(y)] \left(\frac{t}{\text{ns}} \right)^{\frac{7}{192}} \text{ km/s}$$

where:

$$u_{\text{left}}(y) = -247.47183 + 979.34686y^{0.69118} - 1480.07779y^{20.69118} - 1480.07779y^{30.69118} - 1480.07779y^{40.69118}$$

$$u_{\text{right}}(y) = -0.00048 - 104.02827(1-y) + 301.00969(1-y)^2 - 592.03112(1-y)^3 + 437.02875(1-y)^4$$

$$\text{with } W(y) = \frac{1-y}{1+(y/0.20621)^4}.$$

2.2 Specific Volume $v(m, t)$

1. **Dimensionless Fit Function:**

$$V(y) \approx 483.36492 (1 - y)^{39.54884}$$

2. **Theoretical Scaling:**

$$v(m, t) = V(y) A^{a_v} B^{b_v} t^{c_v} = V(y) A^{-\frac{115}{96}} B^{-\frac{25}{48}} t^{\frac{25}{48}}$$

3. **Parametrized Shape:**

$$v(m, t) \approx 483.36492 (1 - y)^{39.54884} A^{-\frac{115}{96}} B^{-\frac{25}{48}} t^{\frac{25}{48}}$$

4. **Substitution of scaling constants and units:** Using physical conversions ($v_{\text{scale}} \approx 1.24268 \text{ cm}^3/\text{g}$):

$$v(m, t) \approx 483.36492 (1 - y)^{39.54884} \cdot 1.24268 \left(\frac{t}{\text{ns}} \right)^{\frac{25}{48}} \text{ cm}^3/\text{g}$$

$$\approx 600.66640 (1 - y)^{39.54884} \left(\frac{t}{\text{ns}} \right)^{\frac{25}{48}} \text{ cm}^3/\text{g}$$

5. **Fully Substituted CGS function:** Substituting $y \approx 981.45247 \left(\frac{m}{\text{g/cm}^2} \right) \left(\frac{t}{\text{ns}} \right)^{-\frac{33}{64}}$:

$$v(m, t) \approx 600.66640 \left(1 - 981.45247 \left(\frac{m}{\text{g/cm}^2} \right) \left(\frac{t}{\text{ns}} \right)^{-\frac{33}{64}} \right)^{39.54884} \left(\frac{t}{\text{ns}} \right)^{\frac{25}{48}} \text{ cm}^3/\text{g}$$

2.3 Pressure $p(m, t)$

1. **Dimensionless Fit Function:** Using the optimal Candidate 4 (Power Law $y = 1$ 3-Term 6-Parameter model):

$$P(y) \approx 0.38708 - 0.18660(1 - y)^{0.49607} - 0.18444(1 - y)^{1.51128} - 0.01335(1 - y)^{6.20384}$$

2. **Theoretical Scaling:**

$$p(m, t) = P(y) A^{a_p} B^{b_p} t^{c_p} = P(y) A^{\frac{505}{192}} B^{\frac{43}{96}} t^{-\frac{43}{96}}$$

3. **Parametrized Shape:**

$$p(m, t) \approx P(y) A^{\frac{505}{192}} B^{\frac{43}{96}} t^{-\frac{43}{96}}$$

4. **Substitution of scaling constants and units:** Using physical conversions ($p_{\text{scale}} \approx 7.05133$ MBar):

$$p(m, t) \approx P(y) \cdot 7.05133 \left(\frac{t}{\text{ns}} \right)^{-\frac{43}{96}} \text{ MBar}$$

5. **Fully Substituted CGS function:** Substituting $y \approx 981.45247 \left(\frac{m}{\text{g/cm}^2} \right) \left(\frac{t}{\text{ns}} \right)^{-\frac{33}{64}}$:

$$p(m, t) \approx [2.72945 - 1.31578(1 - y)^{0.49607} - 1.30057(1 - y)^{1.51128} - 0.09415(1 - y)^{6.20384}] \left(\frac{t}{\text{ns}} \right)^{-\frac{43}{96}} \text{ MBar}$$

2.4 Temperature $T(m, t)$

1. **Dimensionless Fit Function:**

$$T(y) \approx [(1 - y)(1 + 0.20287y)]^{\frac{10}{39}}$$

2. **Theoretical Scaling:**

$$T(m, t) = T(y) T_0$$

3. **Parametrized Shape:**

$$T(m, t) \approx [(1 - y)(1 + 0.20287y)]^{\frac{10}{39}} T_0$$

4. **Substitution of scaling constants and units:** Using physical boundary drive ($T_0 = 1.00000$ HeV):

$$T(m, t) \approx 1.00000 [(1 - y)(1 + 0.20287y)]^{\frac{10}{39}} \text{ HeV}$$

5. **Fully Substituted CGS function:** Substituting $y \approx 981.45247 \left(\frac{m}{\text{g/cm}^2} \right) \left(\frac{t}{\text{ns}} \right)^{-\frac{33}{64}}$:

$$T(m, t) \approx 1.00000 \left[\left(1 - 981.45247 \left(\frac{m}{\text{g/cm}^2} \right) \left(\frac{t}{\text{ns}} \right)^{-\frac{33}{64}} \right) \left(1 + 199.10933 \left(\frac{m}{\text{g/cm}^2} \right) \left(\frac{t}{\text{ns}} \right)^{-\frac{33}{64}} \right) \right]^{\frac{10}{39}} \text{ HeV}$$

2.5 Density $\rho(m, t)$

Note: The subsonic density profile diverges sharply at the ablation front ($\xi \rightarrow 1$), so instead of independently fitting $\tilde{\rho}(\xi)$, we compute it explicitly using the equation of state (EOS) along with the well-behaved analytical fits for pressure $p(m, t)$ and temperature $T(m, t)$.

1. **Theoretical relation:**

$$\rho(m, t) = \left(\frac{r f T(m, t)^\beta}{p(m, t)} \right)^{\frac{1}{\mu-1}}$$

2. **CGS Parametrized form:**

$$\rho(m, t) = \left(\frac{r f_{\text{Kelvin}} T(m, t)^\beta}{p(m, t)} \right)^{\frac{1}{\mu-1}}$$

3. **Fully Substituted CGS function:** Substituting the actual dimensional fits of $T(m, t)$ and $p(m, t)$ (where $r = 0.25$, $\beta = 1.6$, $\mu = 0.14$) yields the final explicit solution:

$$\rho(m, t) \approx 4.03377 \cdot \left[\frac{T(m, t)^{1.6}}{p(m, t)} \right]^{-1.16279} \text{ g/cm}^3$$

3 Explicit Shock compressed Profiles ($m_f(t) < m < m_s(t)$)

3.1 Fluid Velocity $u(m, t)$

1. **Dimensionless Fit Function:** Using the optimal Candidate 3 (3-Parameter Power Law):

$$U(y_s) \approx 0.96888 + 0.18376 y_s^{0.59966}$$

2. **Theoretical Scaling:**

$$u(m, t) = U(y_s) v_0^{\frac{1}{2}} p_0^{\frac{1}{2}} t^{\frac{\tau}{2}}$$

3. **Parametrized Shape:**

$$u(m, t) \approx U(y_s) v_0^{\frac{1}{2}} p_0^{\frac{1}{2}} t^{\frac{\tau}{2}}$$

4. **Substitution of scaling constants and units:** Using physical conversions ($u_{\text{scale}} \approx 3.74525 \text{ km/s}$):

$$u(m, t) \approx U(y_s) \cdot 3.74525 \left(\frac{t}{\text{ns}} \right)^{-\frac{43}{192}} \text{ km/s}$$

5. **Fully Substituted CGS function:** Substituting $y_s \approx 82.68924 \left(\frac{m}{\text{g/cm}^2} \right) \left(\frac{t}{\text{ns}} \right)^{-\frac{149}{192}}$:

$$u(m, t) \approx \left[3.62869 + 0.68824 (y_s)^{0.59966} \right] \left(\frac{t}{\text{ns}} \right)^{-\frac{43}{192}} \text{ km/s}$$

3.2 Pressure $p(m, t)$

1. Dimensionless Fit Function:

$$P(y_s) \approx 1.0 + 0.49535 y_s^{1.13315}$$

2. Theoretical Scaling:

$$p(m, t) = P(y_s) p_0 t^\tau$$

3. Parametrized Shape:

$$p(m, t) \approx P(y_s) p_0 t^\tau$$

4. Substitution of scaling constants and units: Using physical conversions ($p_0 = 2.71000$ MBar):

$$p(m, t) \approx 2.71000 P(y_s) \left(\frac{t}{\text{ns}} \right)^{-\frac{43}{96}} \text{ MBar}$$

5. Fully Substituted CGS function: Substituting $y_s \approx 82.68924 \left(\frac{m}{\text{g/cm}^2} \right) \left(\frac{t}{\text{ns}} \right)^{-\frac{149}{192}}$:

$$p(m, t) \approx [2.71000 + 1.34240 y_s^{1.13315}] \left(\frac{t}{\text{ns}} \right)^{-\frac{43}{96}} \text{ MBar}$$

3.3 Density $\rho(m, t)$

In the shock compressed region under the `FIT_RHO_AROUND_PISTON` option, the density profile is piecewise. Near the piston ($y_s < 0.1$), the density is fitted directly to capture the high-compression boundary behavior. Near the shock front ($y_s \geq 0.1$), the density is derived algebraically from the equation of state (EOS) using the fitted temperature and pressure profiles to ensure thermodynamic consistency.

1. Dimensionless Fit Function:

$$\rho(y_s) \approx \begin{cases} \rho_s + (\rho_0 - \rho_s)(1 - y_s^{0.45561})^{0.69201} & \text{for } y_s < 0.1 \\ \left(\frac{P(y_s)}{r f_{\text{dim}} T(y_s)^\beta} \right)^{\frac{1}{1-\mu}} & \text{for } y_s \geq 0.1 \end{cases}$$

2. Theoretical Scaling:

$$\rho(m, t) = \frac{1}{v_0} \rho(y_s) = \rho_0 \rho(y_s)$$

3. Substitution of scaling constants and units: Using physical density $\rho_0 = 19.32$ g/cm³:

$$\rho(m, t) \approx 19.32 \rho(y_s) \text{ g/cm}^3$$

4. Fully Substituted CGS function: Substituting $y_s \approx 82.68924 \left(\frac{m}{\text{g/cm}^2} \right) \left(\frac{t}{\text{ns}} \right)^{-\frac{149}{192}}$:

$$\rho(m, t) \approx \begin{cases} 9.00000 + (0.00012 - 9.00000)(1 - y_s^{0.45561})^{0.69201} & \text{for } y_s < 0.1 \\ \left(\frac{p(m, t)}{r f_{\text{Kelvin}} T(m, t)^\beta} \right)^{\frac{1}{1-\mu}} & \text{for } y_s \geq 0.1 \end{cases} \text{ g/cm}^3$$

3.4 Temperature $T(m, t)$

Complementary to the density profile, the temperature is piecewise. Near the piston ($y_s < 0.1$), the temperature is derived algebraically from the EOS. Near the front ($y_s \geq 0.1$), the temperature is fitted directly using the optimal Candidate 2 (Double Power Law model) to capture the thermal structure accurately.

1. Dimensionless Fit Function:

$$T(y_s) \approx \begin{cases} \left(\frac{P(y_s) \rho(y_s)^{\mu-1}}{r f_{\text{dim}}} \right)^{\frac{1}{\beta}} & \text{for } y_s < 0.1 \\ T_s - 6.82764(1 - y_s)^{-1.06205} + 8.03503(1 - y_s)^{0.99688} & \text{for } y_s \geq 0.1 \end{cases}$$

2. CGS Parametrized form (derived from CGS EOS in Kelvin):

$$T(m, t) \approx \begin{cases} \left(\frac{p(m, t) \rho(m, t)^{\mu-1}}{r f_{\text{Kelvin}}} \right)^{\frac{1}{\beta}} & \text{for } y_s < 0.1 \\ T_{\text{scale_Kelvin}} [T_s - 6.82764(1 - y_s)^{-1.06205} + 8.03503(1 - y_s)^{0.99688}] \left(\frac{t}{\text{ns}} \right)^{T_{\text{pow}}} & \text{for } y_s \geq 0.1 \end{cases} \quad \text{Kelv}$$

3. Fully Substituted CGS function: Substituting the fitted coefficients (where $T_s = 0.00380$):

$$T(m, t) \approx \begin{cases} \left(\frac{p(m, t) \rho(m, t)^{-0.86}}{0.25 \cdot 6730.91} \right)^{0.625} & \text{for } y_s < 0.1 \\ 115642.73611 [0.00380 - 6.82764(1 - y_s)^{-1.06205} + 8.03503(1 - y_s)^{0.99688}] \left(\frac{t}{\text{ns}} \right)^{0.13717} & \text{for } y_s \geq 0.1 \end{cases}$$

4 Boxed Summary of Piecewise Profiles

4.1 Subsonic Ablation Region ($m < m_f(t)$)

$$m_f(t) \approx 0.00102 \left(\frac{t}{\text{ns}} \right)^{\frac{33}{64}} \text{ g/cm}^2 \quad (28)$$

$$y = 981.45247 \left(\frac{m}{\text{g/cm}^2} \right) \left(\frac{t}{\text{ns}} \right)^{-\frac{33}{64}} \quad (29)$$

$$u(m, t) \approx [W(y)u_{\text{left}}(y) + (1 - W(y))u_{\text{right}}(y)] \left(\frac{t}{\text{ns}} \right)^{\frac{7}{192}} \text{ km/s} \quad (30)$$

where:
$$\begin{cases} u_{\text{left}}(y) = -247.47183 + 979.34686y^{0.69118} - 1480.07779y^{20.69118} - 1480.07779y^{30.69118} - 1480.07779y^{40.69118} \\ u_{\text{right}}(y) = -0.00048 - 104.02827(1 - y) + 301.00969(1 - y)^2 - 592.03112(1 - y)^3 + 437.02875(1 - y)^4 \\ W(y) = \frac{1-y}{1+(y/0.20621)^4} \end{cases}$$

$$v(m, t) \approx 600.66640 \left(1 - 981.45247 \left(\frac{m}{\text{g/cm}^2} \right) \left(\frac{t}{\text{ns}} \right)^{-\frac{33}{64}} \right)^{39.54884} \left(\frac{t}{\text{ns}} \right)^{\frac{25}{48}} \text{ cm}^3/\text{g} \quad (31)$$

$$p(m, t) \approx [2.72945 - 1.31578(1 - y)^{0.49607} - 1.30057(1 - y)^{1.51128} - 0.09415(1 - y)^{6.20384}] \left(\frac{t}{\text{ns}} \right)^{-\frac{43}{96}} \text{ MBar} \quad (32)$$

$$T(m, t) \approx 1.00000 \left[\left(1 - 981.45247 \left(\frac{m}{\text{g/cm}^2} \right) \left(\frac{t}{\text{ns}} \right)^{-\frac{33}{64}} \right) \left(1 + 199.10933 \left(\frac{m}{\text{g/cm}^2} \right) \left(\frac{t}{\text{ns}} \right)^{-\frac{33}{64}} \right) \right]^{\frac{10}{39}} \text{ eV} \quad (33)$$

$$\rho(m, t) \approx 4.03377 \cdot \left[\frac{T(m, t)^{1.6}}{p(m, t)} \right]^{-1.16279} \text{ g/cm}^3 \quad (34)$$

4.2 Shock Compressed Region ($m_f(t) \leq m < m_s(t)$)

$$m_s(t) \approx 0.01209 \left(\frac{t}{\text{ns}} \right)^{\frac{149}{192}} \text{ g/cm}^2 \quad (35)$$

$$y_s = 82.68924 \left(\frac{m}{\text{g/cm}^2} \right) \left(\frac{t}{\text{ns}} \right)^{-\frac{149}{192}} \quad (36)$$

$$u(m, t) \approx \left[3.62869 + 0.68824 (y_s)^{0.59966} \right] \left(\frac{t}{\text{ns}} \right)^{-\frac{43}{192}} \text{ km/s} \quad (37)$$

$$p(m, t) \approx \left[2.71000 + 1.34240 y_s^{1.13315} \right] \left(\frac{t}{\text{ns}} \right)^{-\frac{43}{96}} \text{ MBar} \quad (38)$$

$$\rho(m, t) \approx \begin{cases} 9.00000 + (0.00012 - 9.00000)(1 - y_s^{0.45561})^{0.69201} & \text{for } y_s < 0.1 \\ \left(\frac{p(m, t)}{0.25 \cdot 6730.91 \cdot T(m, t)^{1.6}} \right)^{-1.16279} & \text{for } y_s \geq 0.1 \end{cases} \text{ g/cm}^3 \quad (39)$$

$$T(m, t) \approx \begin{cases} \left(\frac{p(m, t) \rho(m, t)^{-0.86}}{0.25 \cdot 6730.91} \right)^{0.625} & \text{for } y_s < 0.1 \\ 115642.73611 \left[0.00380 - 6.82764(1 - y_s)^{-1.06205} + 8.03503(1 - y_s)^{0.99688} \right] \left(\frac{t}{\text{ns}} \right)^{0.13717} & \text{for } y_s \geq 0.1 \end{cases} \quad (40)$$